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Thermodynamic description of the Dy–Ni system

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ABSTRACT

The Dy–Ni binary system has been thermodynamically assessed by means of the computer program Thermo-Calc. The Redlich–Kister polynomial was used to describe the solution phase, liquid (L). Ten compounds, Dy_3Ni, Dy_3Ni_2, DyNi, $DyNi_2$, $DyNi_3$, Dy_2Ni_7, $DyNi_4$, Dy_4Ni_{17}, $DyNi_5$ and Dy_2Ni_7, were treated as stoichiometric phases. The parameters of the Gibbs energy expressions were optimized according to all the available experimental information of both the equilibrium data and the thermodynamic results. A set of self-consistent thermodynamic parameters of the Dy–Ni system has been obtained. The calculations agree well with the respective experimental data.

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1. Introduction

In recent years, rare earth transition metal alloys have attracted considerable practical interest due to their functional properties, i.e. their excellent magnetic properties, hydrogen absorbing or permeable properties, and catalytic properties [1–4]. Considering a systematic design and processing of the materials, the knowledge of phase diagrams and the thermodynamic properties is of great importance. Binary thermodynamic information forms the basis of a powerful approach to the phase stability of multicomponent alloy systems using the CALPHAD [5] technique. Phase diagram calculation, using thermodynamic models coupled with key experimental measurements, is an efficient way to obtain the phase diagram.

The Dy–Ni system has not been assessed thermodynamically. The purpose of this work is to carry out thermodynamic assessments of the Dy–Ni system by means of the CALPHAD (calculation of phase diagrams) method, in which the Gibbs free energy of a phase is described by a thermodynamic model. The thermodynamic parameters of the Dy–Ni system are optimized by fitting the experimental data to thermodynamic properties and phase equilibria.

2. Experimental information

2.1. Experimental phase diagram data

The Dy–Ni system was first investigated by Nassau and Wallace [6] who indicated the existence of two compounds $DyNi_2$ and $DyNi_5$ and speculated the existence of the phases Dy_5Ni and DyNi. Baenziger and Moriarty [7] studied the Dy–Ni system and also confirmed the existence of two compounds $DyNi_2$ and $DyNi_5$. The two intermediate phases Dy_3Ni and DyNi were detected. They suggested that two of the three compounds $DyNi_3$, Dy_2Ni_7 and $DyNi_4$ could exist. Copeland and Kato [8] studied the Dy-rich side of the phase diagram. The intermetallic compound of Dy_3Ni is formed by peritectic reaction at 33 at.% Ni. Many researchers studied the intermediate phases and confirmed the existence of eight compounds: Dy_3Ni, Dy_3Ni_2, DyNi, $DyNi_2$, $DyNi_3$, Dy_2Ni_7, $DyNi_5$ and Dy_2Ni_{17} [9–13].

The entire phase diagram of Dy–Ni was investigated by Zheng and Wang [14] using X-ray in conjunction with differential thermal analysis. The intermetallic compound $DyNi_4$ was detected, and the new intermetallic compound Dy_4Ni_{17} was observed. In this system, there were ten intermetallic compounds: Dy_3Ni, Dy_3Ni_2, DyNi, $DyNi_2$, $DyNi_3$, Dy_2Ni_7, $DyNi_4$, Dy_4Ni_7, $DyNi_3$ and Dy_2Ni_{17}. The intermetallic compounds DyNi and $DyNi_5$ melted congruently at 1521 K and 1660 K, and the intermetallic compounds Dy_3Ni, Dy_3Ni_2, DyNi, $DyNi_3$, Dy_2Ni_7, $DyNi_4$, and Dy_3Ni_{17} were formed by peritectic reactions at 1035 K, 1201 K, 1531 K, 1580 K, 1609 K, 1625 K and 1594 K, respectively. Three eutectic reactions occurred: at 31 at.% Ni and 966 K, 56 at.% Ni and 1446 K, 93 at.% Ni and 1552 K. Neither Dy in Ni nor Ni in Dy showed any detectable solid solubility [14].

The temperature and composition of the invariant reactions in the Dy–Ni system are listed in Table 1.

2.2. Thermodynamic data

Many researchers studied the thermodynamic properties of Dy–Ni intermetallic compounds. Tong et al. [15] measured the

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Table 1
The temperatures and the compositions of the invariant reactions in the Dy–Ni system.

Invariant reaction	Composition of phases at.% Ni			Temperature (K)	Reaction type	Ref.
$L + (\alpha Dy) \rightarrow Dy_3Ni$	0.275	0	0.25	1035	Peritectic	[6]
	0.251	0	0.25	1032	Peritectic	Optimized
$L \rightarrow Dy_3Ni_2 + Dy_3Ni$	0.31	0.4	0.25	966	Eutectic	[6]
	0.30	0.4	0.25	970	Eutectic	Optimized
$L + DyNi \rightarrow Dy_3Ni_2$	0.34	0.4	0.3	1201	Peritectic	[6]
	0.37	0.4	0.3	1200	Peritectic	Optimized
$L \rightarrow DyNi$	0.5	0.5		1521	Congruent	[6]
	0.5	0.5		1527	Congruent	Optimized
$L \rightarrow DyNi + DyNi_2$	0.56	0.5	0.667	1446	Eutectic	[6]
	0.567	0.5	0.667	1434	Eutectic	Optimized
$L + DyNi_3 \rightarrow DyNi_2$	0.61	0.75	0.667	1531	Peritectic	[6]
$L \rightarrow DyNi_3 + DyNi_2$	0.70	0.75	0.667	1540	Eutectic	Optimized
$L \rightarrow DyNi_2$	0.667	0.667		1558	Congruent	Optimized
$L + Dy_2Ni_7 \rightarrow DyNi_3$	0.628	0.778	0.75	1556	Peritectic	[6]
	0.721	0.778	0.75	1554	Peritectic	Optimized
$L + DyNi_4 \rightarrow Dy_2Ni_7$	0.646	0.8	0.778	1580	Peritectic	[6]
	0.741	0.8	0.778	1579	Peritectic	Optimized
$L + Dy_4Ni_{17} \rightarrow DyNi_4$	0.69	0.81	0.8	1609	Peritectic	[6]
	0.763	0.81	0.8	1608	Peritectic	Optimized
$L + DyNi_5 \rightarrow Dy_4Ni_{17}$	0.69	0.833	0.81	1625	Peritectic	[6]
	0.779	0.833	0.81	1625	Peritectic	Optimized
$L \rightarrow DyNi_5$	0.833	0.833		1670	Congruent	[6]
	0.833	0.833		1669	Congruent	Optimized
$L + DyNi_5 \rightarrow Dy_2Ni_{17}$	0.911	0.833	0.894	1594	Peritectic	[6]
	0.906	0.833	0.894	1586	Peritectic	Optimized
$L \rightarrow (Ni) + Dy_2Ni_{17}$	0.93	1	0.894	1552	Eutectic	[6]
	0.934	1	0.894	1565	Eutectic	Optimized

standard Gibbs free energy of formation for $DyNi_5$ obtained by EMF measurements in a molten NaCl–KCl–$DyCl_3$ system at 973 K. Schott and Sommer [16], Guo and Kleppa [17] and Deodhar and Ficalora [18] researched the standard enthalpies of formation for several Dy–Ni intermetallic compounds by the calorimetric measurements at 1098, 1474 K and 985 K, respectively.

Electromotive force (EMF) measurements for various Dy–Ni intermetallic compounds in two-phase coexisting states were carried out in the temperature range of 673–773 K in a molten LiCl–KCl–$DyCl_3$ system by Konishi et al. [19]. The activities and relative partial molar Gibbs free energies of Dy were obtained from the measured EMFs for various Dy–Ni intermetallic compounds, $DyNi_2$, $DyNi_3$, Dy_2Ni_7, and $DyNi_5$. The relative partial molar entropies and enthalpies of Dy were also calculated from the temperature dependence of the EMFs. The activities and relative partial molar properties of Ni in the compounds were calculated from the activities of Dy by using the Gibbs–Duhem equation. Meanwhile, the standard Gibbs free energies of formation for the Dy–Ni intermetallic compounds were estimated [19].

3. Thermodynamic models

3.1. Pure elements

The stable forms of the pure elements at 298.15 K were chosen as the reference states of the system. For the thermodynamic functions of the pure elements in their stable and metastable states, the phase stability equations compiled by Dinsdale [20] were used in the present optimization. The equations are of the SGTE (Scientific Group Thermodata Europe) format:

$$\begin{aligned} {}^0G_i^{\phi}(T) &= G_i^{\phi}(T) - H_i^{SER}(298.15\ \mathrm{K}) \\ &= a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1) \end{aligned}$$

where H_i^{SER} (298.15 K) is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, *hcp* for Dy and *fcc* for Ni. $G_i^{\phi}(T)$ and ${}^0G_i^{\phi}(T)$ are the absolute and the relative Gibbs energy of the element i in the ϕ state. The Gibbs energy of the element i (i = Dy and Ni), ${}^0G_i^{\phi}(T)$, in its SER state is denoted by $GHSER_i$, i.e.

$$GHSER_{Dy} = {}^0G_{Dy}^{hcp}(T) = G_{Dy}^{hcp}(T) - H_{Dy}^{SER}(298.15\ \mathrm{K}) \quad (2)$$

$$GHSER_{Ni} = {}^0G_{Ni}^{fcc}(T) = G_{Ni}^{fcc}(T) - H_{Ni}^{SER}(298.15\ \mathrm{K}) \quad (3)$$

3.2. Solution phases

In the Dy–Ni system, The solution phase, liquid, was described by the substitutional solution model. The Gibbs energy function of the solution phase ϕ for 1 mole of atoms is described by the following expression:

$$G_m^{Liq} = x_{Dy}\,{}^0G_{Dy}^{Liq} + x_{Ni}\,{}^0G_{Ni}^{Liq} + RT(x_{Dy}\ln x_{Dy} + x_{Ni}\ln x_{Ni}) + {}^EG_m^{Liq} \quad (4)$$

where ${}^0G_i^{Liq}$ is the molar Gibbs energy of the element i (i = Dy and Ni) with the structure of liquid. ${}^EG_m^{Liq}$ is the excess Gibbs energy which is expressed in the Redlich–Kister polynomial:

$${}^EG_m^{Liq} = x_{Dy}x_{Ni}\sum_i {}^iL^{\phi}(x_{Dy} - x_{Ni})^i. \quad (5)$$

The parameter ${}^iL^{Liq}$ ($i = 0, 1, 2, \ldots$) is the ith interaction parameter between the elements Dy and Ni and to be evaluated in the present work. Its general form is as the following:

$${}^iL^{\phi} = a_i + b_iT + c_iT\ln T + d_iT^2 + e_iT^3 + f_iT^{-1} + g_iT^7 + h_iT^{-9} \quad (6)$$

where $a_i, b_i, c_i, d_i, e_i, f_i, g_i$ and h_i are the coefficients to be optimized. In most cases, only the first one or two terms of the above equation are used.

3.3. Stoichiometric compounds

In the present optimization, Dy_3Ni, Dy_3Ni_2, DyNi, $DyNi_2$, $DyNi_3$, Dy_2Ni_7, $DyNi_4$, Dy_4Ni_{17}, $DyNi_5$ and Dy_2Ni_7, phases are treated as the stoichiometric compounds. The Gibbs energy per mole of formula unit Dy_mNi_n is expressed as follows:

$$\begin{aligned} {}^0G_m^{Dy_mNi_n} &= G_m^{Dy_mNi_n} - mH_{Dy}^{SER} - nH_{Ni}^{SER} \\ &= m\,{}^0G_{Dy}^{hcp} + n\,{}^0G_{Ni}^{fcc} + \Delta G_f^{Dy_mNi_n} \quad (7) \end{aligned}$$

where $\Delta G_f^{Dy_mNi_n}$ is the Gibbs energy of formation for per mole of formula unit Dy_mNi_n. $\Delta G_f^{Dy_mNi_n}$ can be given by the following expression:

$$\Delta G_f^{Dy_mNi_n} = a + bT \quad (8)$$

where a and b are the parameters to be evaluated in the present work.

4. Assessment procedure

The thermodynamic optimization of the model parameters of the Gibbs energy expressions is an application of the CALPHAD technique with the help of the PARROT module of the Thermo-Calc software developed by Jansson [21] and Sundman et al. [22]. Its procedure consists of the choice of thermodynamic models for the Gibbs energy of the individual phases as previously described, the analysis of all the related experimental data available, and the computer-aided nonlinear regression for minimizing the square sum of the errors between the experimental data and the computed values.

The experimental data from Zheng and Wang [14] were used in the optimization. The data from [14] were offered relatively large weight factors since they had constructed a reasonable phase diagram over almost the entire composition range. In this study, the thermodynamic parameters for the intermetallic compounds were optimized at the first stage based on the standard enthalpies of formation for the Dy–Ni intermetallic compounds [16–19,23] and the phase boundary information [14]. The data [16–18] were offered relatively large weight factors since calorimetric measurements of the enthalpy of formation are generally more reliable than extrapolations from emf measurements.

In the beginning of the assessment, each item of the selected information was given a certain weight factor by judgment. During the period of optimization, the weight factors were adjusted by trial and error. The final data set was obtained when the squared sum of the errors between the experimental data and the calculated results was reduced to a certain level.

5. Results and discussion

The thermodynamic description and the optimized parameters of the Dy–Ni system are listed in Table 2. The phase diagram calculated by the present parameters is shown in Fig. 1. The calculated results are compared with the experimental data in Fig. 2 and are nearly identical to the one represented by Zheng and Wang [14]. The main differences concern the melting behavior of Dy_3Ni and $DyNi_2$. In the present work, the phase $DyNi_2$ melted congruently and eutectic reactions L $\rightarrow$ $DyNi_2$ + $DyNi_3$ are obtained rather than the peritectic reaction L + $DyNi_3$ $\rightarrow$ $DyNi_2$ reported by Zheng and Wang [14]. The incompatible asymmetry of the liquidus on the left and right sides of the compounds $DyNi_5$ is not maintained.

All invariant equilibria in the system given by Zheng and Wang [14] are reproduced, which are listed in Table 1.

The compositions and temperatures of invariant reactions fit well with the experiments [14] except for the eutectic reaction, L $\rightarrow$ (Ni) + Dy_2Ni_{17}, for which the calculated temperature is 13 K higher than the experimental data. The calculated standard enthalpies of formation of the Dy–Ni intermediate phases at 700 K with experimental measurements [16–19] and the present predicted values using the Miedema model [23] are shown in Fig. 3. The calculated standard Gibbs free energies of formation for the Dy–Ni intermetallic compounds at 700 K and the experimental measurements from [19] are shown in Fig. 4. Fig. 5 shows the calculated activities of Dy and Ni in the Dy–Ni system compared with the experimental data from [19]. Within the experimental uncertainties, most of the experimental data are well reproduced by the present calculation.

Table 2
The thermodynamic parameters of Dy–Ni system.[a]

Phase	Parameters
Liquid	${}^0L^{Liquid} = -185784.1 + 33.570 \times T$
	${}^1L^{Liquid} = 17309 + 11.393 \times T$
	${}^2L^{Liquid} = 37829.7 - 9.932 \times T$
Dy_3Ni	${}^0G_m^{Dy_3Ni} = 3{}^0G_{Dy}^{hcp} + {}^0G_{Ni}^{fcc} - 92416.2 - 1.596 \times T$
Dy_3Ni_2	${}^0G_m^{Dy_3Ni_2} = 3{}^0G_{Dy}^{hcp} + 2{}^0G_{Ni}^{fcc} - 180610.3 + 2.140 \times T$
DyNi	${}^0G_m^{DyNi} = {}^0G_{Dy}^{hcp} + {}^0G_{Ni}^{fcc} - 89521.8 + 5.207 \times T$
$DyNi_2$	${}^0G_m^{DyNi_2} = {}^0G_{Dy}^{hcp} + 2{}^0G_{Ni}^{fcc} - 152030.4 + 23.450T$
$DyNi_3$	${}^0G_m^{DyNi_3} = {}^0G_{Dy}^{hcp} + 3{}^0G_{Ni}^{fcc} - 177245.6 + 28.677 \times T$
Dy_2Ni_7	${}^0G_m^{Dy_2Ni_7} = 2{}^0G_{Dy}^{hcp} + 7{}^0G_{Ni}^{fcc} - 370138.0 + 57.414 \times T$
$DyNi_4$	${}^0G_m^{DyNi_4} = {}^0G_{Dy}^{hcp} + 4{}^0G_{Ni}^{fcc} - 189767.8 + 27.386 \times T$
Dy_4Ni_{17}	${}^0G_m^{Dy_4Ni_{17}} = 4{}^0G_{Dy}^{hcp} + 17{}^0G_{Ni}^{fcc} - 763294.8 + 104.983 \times T$
$DyNi_5$	${}^0G_m^{DyNi_5} = {}^0G_{Dy}^{hcp} + 5{}^0G_{Ni}^{fcc} - 192403.2 + 22.477 \times T$
Dy_2Ni_{17}	${}^0G_m^{Dy_2Ni_{17}} = 2{}^0G_{Dy}^{hcp} + 17{}^0G_{Ni}^{fcc} - 398025.3 + 48.091 \times T$

[a] In J/(mole of formula units); Temperature(T) in Kelvin.

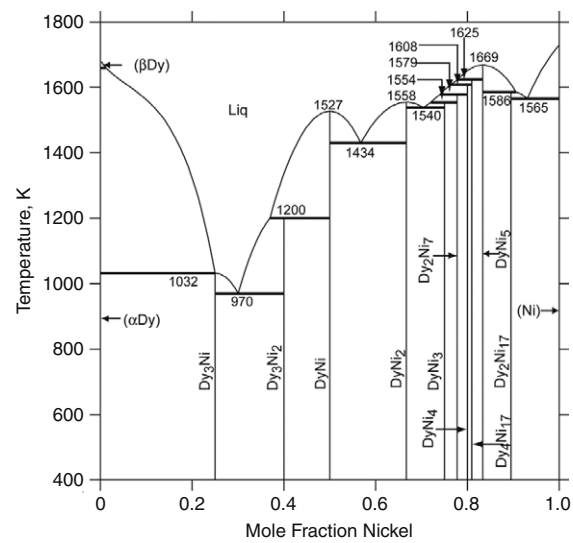

Fig. 1. The Dy–Ni phase diagram calculated from the present thermodynamic description.

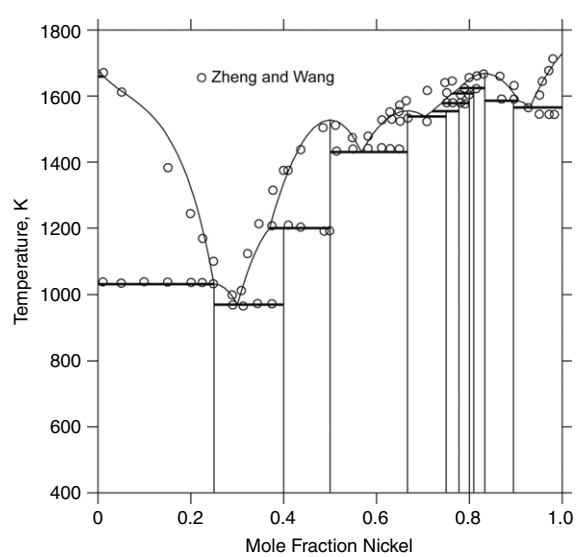

Fig. 2. The Dy–Ni phase diagram calculated from the present thermodynamic description in comparison with experimental measurements [14].

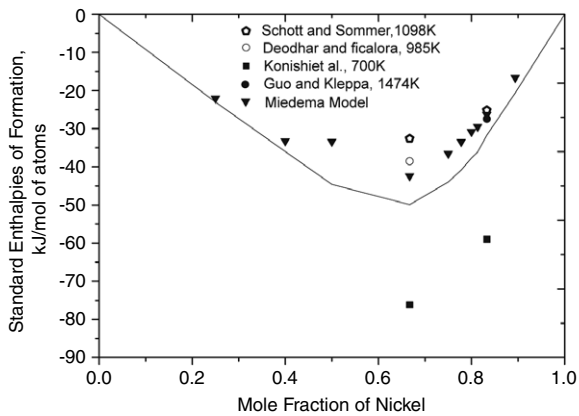

Fig. 3. The calculated standard enthalpies of formation of the Dy–Ni intermediate phases at 700 K with experimental measurements [16–19] and the present predicted values using the Miedema model [23].

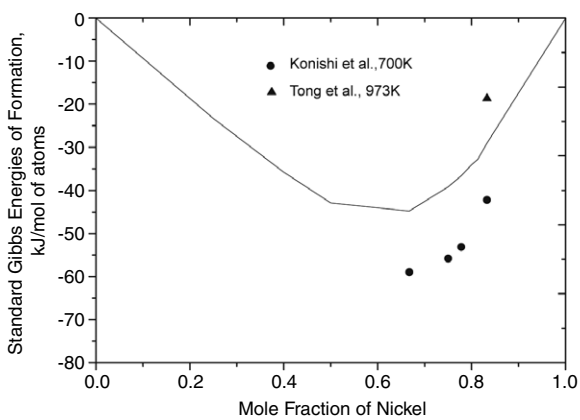

Fig. 4. The calculated standard Gibbs energies of formation of the Dy–Ni intermediate phases at 700 K with experimental measurements [15,19].

6. Conclusions

All of the experimental phase equilibria and thermodynamic data of the Dy–Ni system from the available literature have been critically evaluated. Within the scheme of the CALPHAD technique, the thermodynamic models for all the solution phases and intermediate compounds were selected and the Gibbs energy functions were optimized. A set of consistent thermodynamic parameters for the Dy–Ni binary system was obtained. The calculated phase equilibria and thermodynamic properties, including the phase diagrams, the standard enthalpies and Gibbs free energies of formation for the Dy–Ni intermetallic compounds, and the activities of Dy and Ni, all agree well with the experimental data. The exceptions are the melting behavior of the compounds Dy_3Ni and $DyNi_2$ and their related invariant reactions. The congruent melting and the eutectic reaction are obtained for each of these two compounds instead of the incongruent melting and the peritectic reaction as reported. Detailed experiments are necessary to check these inconsistencies.

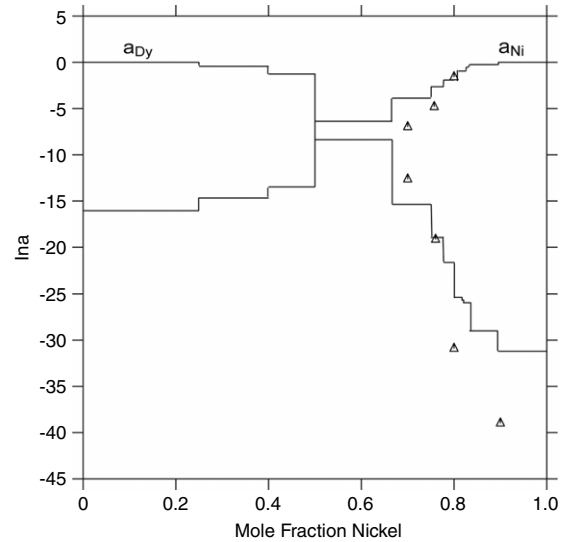

Fig. 5. The calculated activities for Dy–Ni intermetallic compounds in two-phase coexisting states at 773 K with experimental measurements [19].

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